Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 4 Solutions

Problem 4.1

When making investment decisions, it is helpful to group companies by their size and their market. Let random variable $X \in \{1, 2, \dots\}$ represent the company's revenue in millions of dollars, and random variable $Y \in \{\texttt{local}, \texttt{global}\}$ represent whether the company is local (`local`) or global (`global`). For a company chosen at random, the revenue X is a geometric random variable with parameter $1/2$, i.e., $p_X(x) = 1/2^x$ for $x = 1, 2, \dots$. Moreover, the probability that the company has revenue of x million dollars and it is global is $g(x) = (a^x - 1)$ times the probability that the company's revenue is x and it is local. Assume that a is a known constant, $a > 1$.

(a) Determine joint PMF $p_{X,Y}$.

Solution: We marginalize the joint PMF $p_{X,Y}$ with respect to Y to obtain

$$\begin{aligned} p_X(x) &= p_{X,Y}(x, \texttt{local}) + p_{X,Y}(x, \texttt{global}) \\ &= p_{X,Y}(x, \texttt{local}) + (a^x - 1)\, p_{X,Y}(x, \texttt{local}) \qquad (1) \\ &= a^x p_{X,Y}(x, \texttt{local}), \end{aligned}$$

where to obtain (1) we have used that

$$p_{X,Y}(x, \texttt{global}) = (a^x - 1)\, p_{X,Y}(x, \texttt{local})$$

for all $x = 1, 2, \dots$. Since X is a geometric random variable with parameter $1/2$, we obtain

$$p_{X,Y}(x, y) = \begin{cases} \dfrac{1}{(2a)^x} & y = \texttt{local} \\ \dfrac{a^x - 1}{(2a)^x} & y = \texttt{global} \end{cases} \qquad x = 1, 2, \dots$$

(b) Determine the probability that a randomly chosen company is local.

Solution: We marginalize $p_{X,Y}$ with respect to X to obtain

$$\begin{aligned} \mathbb{P}(Y = \texttt{local}) &= p_Y(\texttt{local}) \\ &= \sum_{x=1}^{\infty} p_{X,Y}(x, \texttt{local}) \\ &= \sum_{x=1}^{\infty} \frac{1}{(2a)^x} \\ &= \frac{1}{2a - 1}, \end{aligned}$$

where we have used the geometric series sum formula to obtain the last equality.

Note that $(a^x - 1) \to 0$ and $p_Y(\texttt{local}) \to 1$ as $a \to 1$. Moreover, $(a^x - 1) \to \infty$ and $p_Y(\texttt{local}) \to 0$ as $a \to \infty$, i.e., the probability that a randomly chosen company is local goes to zero. These statements remain true if we restrict our attention to companies with revenue x, i.e., condition on $X = x$ for any $x \in \{1, 2, \dots\}$.

(c) All companies are taxed at a rate $\alpha > 0$ of their revenues. In addition, a global company must pay a tax of $c > 0$ million dollars. Assume α and c are known constants. Determine the expected tax a company chosen at random pays.

Solution:

We define $Z \triangleq \alpha X + c\,\mathbb{1}(Y = \texttt{global})$ to represent the tax a randomly chosen company pays and use the linearity of expectation to obtain

$$\begin{aligned}
\mathbb{E}[Z] &= \mathbb{E}[\alpha X + c\,\mathbb{1}(Y = \texttt{global})] \\
&= \alpha\,\mathbb{E}[X] + c\,\mathbb{E}\big[\mathbb{1}(Y = \texttt{global})\big] \\
&= 2\alpha + c\,p_Y(\texttt{global}) \qquad (2) \\
&= 2\alpha + c\,\big(1 - p_Y(\texttt{local})\big) \\
&= 2\alpha + c\left(1 - \frac{1}{2a-1}\right) \qquad (3) \\
&= 2\alpha + \frac{2(a-1)}{2a-1}\,c,
\end{aligned}$$

where to obtain (2) we have used that the expectation of a geometric random variable with parameter p is $1/p$, and to obtain (3) we have used the result in part (b). Note that the linearity of expectation made it unnecessary to derive the distribution of Z.

Problem 4.2

In this problem, we explore indicator variables.

(a) We toss a fair coin n times. Let X_i be the indicator random variable that corresponds to coin i landing heads. Let Y be the total number of tosses that turned up heads. Determine $\mathbb{E}[X_i]$ and $\mathbb{E}[Y]$.

Solution: Since the coin is fair, $p_{X_i}(1) = p_{X_i}(0) = 1/2$ and

$$\mathbb{E}[X_i] = 1 \cdot p_{X_i}(1) + 0 \cdot p_{X_i}(0) = p_{X_i}(1) = \frac{1}{2}, \qquad i = 1, \dots, n.$$

Note that $\mathbb{E}[S] = p_S(1)$ for any indicator random variable S.

Meanwhile, $Y = \sum_{i=1}^{n} X_i$. The linearity of expectation implies that

$$\mathbb{E}[Y] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \frac{n}{2}.$$

The linearity of expectation enables us to compute expectations of sums and averages of random variables directly, without deriving the distribution of the random variable that represents the sum.

(b) A monkey types 1,000,000,000 characters using the 26 letters of the alphabet. The monkey selects each character independently of all other strokes and is equally likely to choose any of the 26 letters. Determine the expected number of occurrences of the word `caesar`.

Solution: Let X_i be the indicator random variable that the word `caesar` starts at position i. We then have

$$\begin{aligned}
&\mathbb{E}[X_i] \\
&\quad = \mathbb{P}(\text{the word } \texttt{caesar} \text{ starts at position } i) && (4)\\
&\quad = \mathbb{P}(i\text{th letter is } \texttt{c})\,\mathbb{P}((i+1)\text{th letter is } \texttt{a}) \cdots \mathbb{P}((i+5)\text{th letter is } \texttt{r}) && (5)\\
&\quad = \underbrace{\frac{1}{26} \cdot \frac{1}{26} \cdots \frac{1}{26}}_{6 \text{ times}} && (6)\\
&\quad = \frac{1}{26^6} \approx 3.24 \cdot 10^{-9},
\end{aligned}$$

where to obtain (4) we used that $\mathbb{E}[X] = \mathbb{P}(X = 1)$, to obtain (5) we used the independence of strokes, and to obtain (6) we used the fact that the characters are chosen at random.

We define Y to represent the number of occurrences of the word `caesar` and note that $Y = \sum_{i=1}^{999,999,995} X_i$. We use the linearity of expectation to obtain

$$\mathbb{E}[Y] = \mathbb{E}\left[\sum_{i=1}^{999,999,995} X_i\right] = \sum_{i=1}^{999,999,995} \mathbb{E}[X_i] = \frac{999,999,995}{26^6} \approx 3.24.$$

Note that the events $\{X_i = 1\}$ for values of i that are close to each other are not independent. For example, $\mathbb{P}(X_{i+1} = 1 | X_i = 1) = 0$ for $i = 1, \ldots, 999,999,994$. And yet we could evaluate the expectations above by using the linearity of expectation which does not require independence of the underlying events. Looking ahead, the notion of independence can be formalized for random variables as well, as we will see shortly in the course.

(c) Consider a standard 52-card deck that has been shuffled. Determine the expected number of times when a king in the deck is followed by another king. For example,

if all four kings are next to each other, then we'd say that there are 3 kings that are followed by another king.

Solution: We define an indicator random variable X_i that is equal to 1 when the cards in positions i and $i+1$ are both kings. Because shuffling can be thought of as a random permutation of numbers $1, \ldots, 52$, we have

$$\begin{aligned}\mathbb{E}[X_i] &= \mathbb{P}(i\text{th card is a king and } (i+1)\text{th card is a king}) \\ &= \frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} \approx 4.44 \cdot 10^{-3}.\end{aligned}$$

Unlike part (b), here the cards are chosen *without replacement* and therefore the probability of getting the second king in position $i+1$ is different from the probability of getting the first king in position i.

The random variable

$$Y = \sum_{i=1}^{51} X_i$$

represents the number of times a pair of kings appears in the shuffling, whence

$$\mathbb{E}[Y] = \sum_{i=1}^{51} \mathbb{E}[X_i] = 51 \cdot \frac{12}{2652} = \frac{3}{13} \approx 0.23.$$